Characterizing Data Center Development using Multimodal Data Fusion

Abhinav Atthota Swati Thirulogachander Kevin Lam Sajna Bhandari
Advay Chandramouli Dr. Weili Wu

Introduction and Purpose

Rising demand for AI training and inference has accelerated data center construction across the United States, with these facilities accounting for 45.7% of approved development projects. Evaluating their regional energy and water demands remains difficult because public operational data is sparse and conventional audits require manual investigation. This project presents and evaluates a multimodal deep learning framework that integrates public satellite imagery with spatial and environmental indicators (terrain, power infrastructure, and hydrological risk) to classify facility scale.

Data Sourcing & Preprocessing

Dataset: The 2025 IM3 Atlas Dataset (PNNL) provided coordinates and footprints for 1,147 U.S. data centers, used to retrieve and align satellite imagery, USGS elevation data, and regional energy and water indicators. After deduplication, collision removal, and exclusion of small facilities, 173 Medium and Large facilities formed the final dataset.

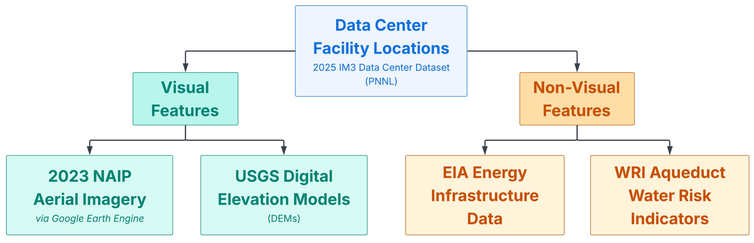


Figure 1. A taxonomy of how visual and non-visual features were aggregated for each facility based on shared coordinates.



Figure 2. Large compute complexes are often split among multiple operators, with overlapping boundaries. Spatial deduplication merges them into single facilities, leaving only distinct facilities in the final dataset.



Figure 3. Sample 2023 NAIP image tiles, which show variance in roof footprint, tree cover and access paths for the model to learn from.

Model

Proposed Architecture: The multimodal model receives three inputs: NAIP imagery (ResNet-18), USGS 3DEP terrain (SmallCNN), and tabular IM3, EIA, and WRI features (MLP). Embeddings are concatenated into a 192-dimensional vector and passed to a linear classifier that predicts facility size class. Explainable AI (XAI) is applied to interpret most predictive attributes in each modality.

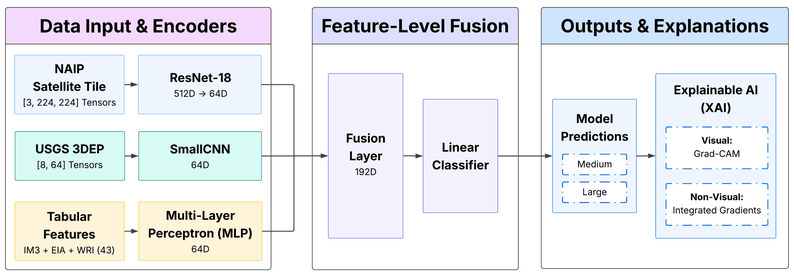


Figure 4. Multimodal architecture: three separate 64D-encoders process imagery (ResNet-18), terrain (SmallCNN), and tabular features (MLP). Their embeddings are concatenated into a 192-dimensional vector and classified as Medium or Large. XAI shows which inputs drive each prediction.

Results

Model Training: Models were trained using class-weighted cross-entropy, AdamW optimization with weight decay, training-only image augmentation, learning-rate scheduling, and early stopping. The pretrained RGB and terrain encoders remained frozen, while modality projections, the tabular encoder, and classification head were trained.

Model Evaluation: Five geographic folds were used to evaluate performance on held-out facilities. The best checkpoint for each fold was selected using validation loss. Results combine one held-out prediction per facility ($n = 173$).

Predictions are evaluated using Macro-F1, which averages per-class F1 scores, and Balanced Accuracy, which averages per-class recall.

Model	Macro-F1 ↑	Balanced Acc. ↑	Accuracy ↑
Image Only (Baseline)	**0.704**	**0.738**	**0.728**
Image + Tabular	0.688	0.726	0.711
Image + Terrain	0.648	0.660	0.694
Image + Tabular + Terrain	0.596	0.623	0.624

Table 1. Four-way modality ablation: modalities are tested individually to the image-only baseline to isolate each one's contribution to Medium/Large classification. Results pool held-out predictions across five geographic folds ($n = 173$).

Analysis

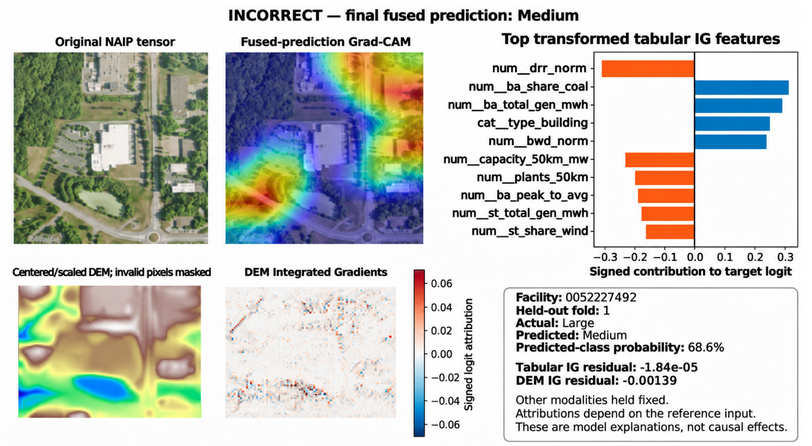


Figure 5. XAI on an incorrect prediction. Top left: Grad-CAM heatmap. Top right: tabular feature attributions (Integrated Gradients). Bottom: terrain attributions (Integrated Gradients).

Facility scale is best captured by RGB imagery. Fusing tabular and terrain data does not improve generalization and instead degrades the visual signal in the multimodal model. In the misclassified example, Grad-CAM focuses on surrounding vegetation, access roads, and development rather than the facility footprint, while the strongest tabular features come from balancing-authority generation and other shared regional indicators. DEM attributions are weak and diffuse, indicating that terrain adds little discriminative value for scale. Together, it suggests resolution mismatch lets coarse scene context regional features dilute the visual signal.

Conclusion

This project is an early step toward scalable data center profiling: public satellite imagery reliably infers facility scale, addressing the lack of standardized reporting in high-growth AI corridors. Naive fusion with coarse regional grid and water indicators exposed a resolution mismatch. Future work should expand facility coverage, fine-tune visual encoders, and test adaptive fusion (e.g., cross-attention) to integrate regional context with local visual features.

Selected References

[1] Zhou, Y., Tan, Y., Wen, Q., Wang, W., Li, L., and Li, Z. (2023). Deep Multimodal Fusion Model for Building Structural Type Recognition Using Multisource Remote Sensing Images and Building-Related Knowledge. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 16, 9646–9660. `https://doi.org/10.1109/JSTARS.2023.3323484`

[2] Suel, E., Bhatt, S., Brauer, M., Flaxman, S., and Ezzati, M. (2021). Multimodal deep learning from satellite and street-level imagery for measuring income, overcrowding, and environmental deprivation in urban areas. *Remote Sensing of Environment*, 257, 112339. `https://doi.org/10.1016/j.rse.2021.112339`

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